

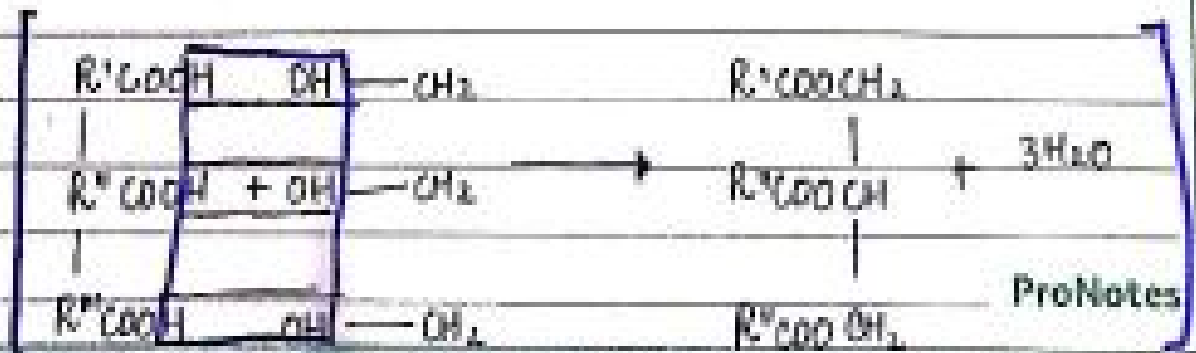
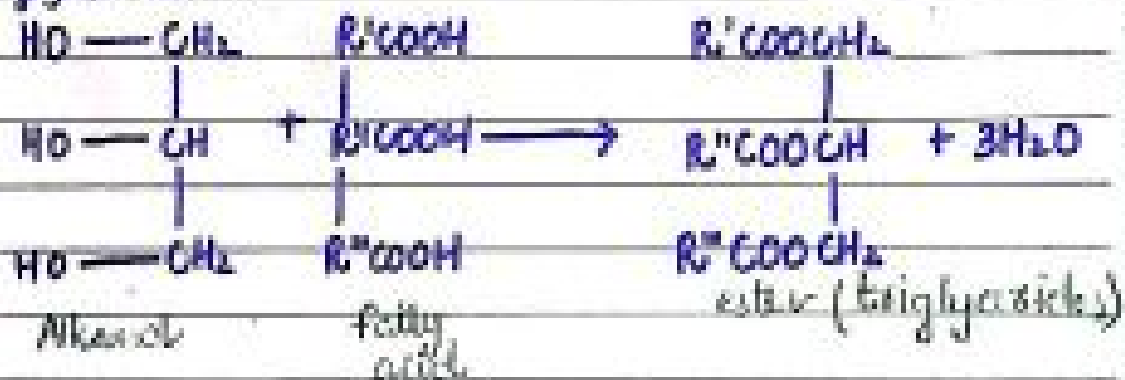
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Assignment

Fats and oils are esters of glycerol and three fatty acids. Fats are also known triglycerides. Oil is any non polar chemical substance that is a viscous liquid at ambient temperatures and is both hydrophobic and lipophilic. Oils have a high carbon and hydrogen content.

Fats and oils belong to a group of compounds known as lipids. Fats are solid and are of animal's origin while oils are liquid mainly from plants.

Triglyceride:



Date _____

Components

Fats and oils are composed of triglycerides, which are esters.

Minor components (1-5%) Major components (95-99%)

Phospholipids

sterols

pigments

vitamins

hydrocarbons

Triglyceride

glycerol

free fatty acids

Mono-diglycerides

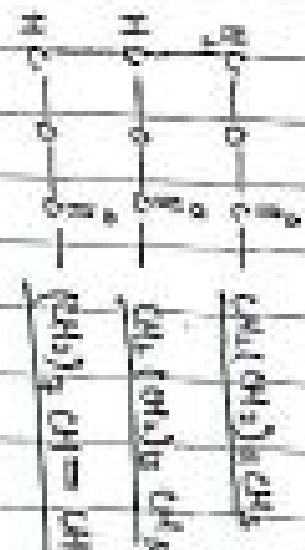
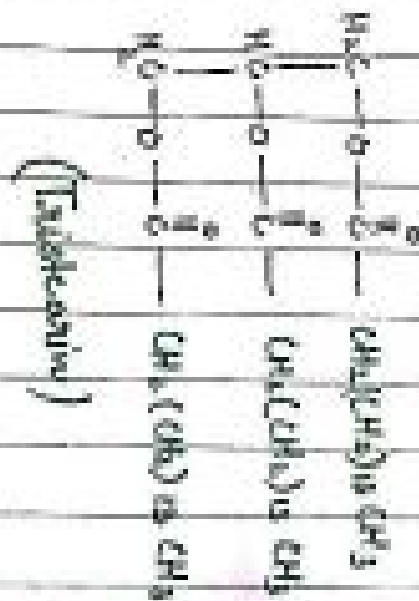
Types of glycerides

simple triglyceride.

If three fatty acids connected to glycerol are of the same type then it is called simple triglyceride.

mixed triglyceride.

If three fatty acids connected to glycerol are of different types it is called mixed triglyceride.



ProNotes

Classification

Fats are classified into saturated and unsaturated fats. This classification is important to enable you to advise your community about which fats can be consumed with less risks to people's health.

As a general rule, plant sources of fats are better for a person's health than the animal sources, because animal fats contain more saturated fats.

Saturated fats

Saturated fats are usually solid at cool temperatures. Eating too much saturated fats is not good for person's health, as it can cause heart and blood vessel problems.

unsaturated fats

Unsaturated fats are usually liquid at room temperature. These types of fats are healthy fats.

e.g.:

oil seeds, maize oil, ground nut oil.

Date _____

Subdivided

<u>Saturated fats</u>	<u>Unsaturated fats</u>	<u>Trans fats</u>
one double bond	two or more double bond	Occur naturally in small amount in some animal product or produced artificially through hydrogenation
<u>Sources</u>	<u>Sources</u>	
• olive oil, avocados, nuts, seeds	• vegetable oils, walnuts, fatty fish	
• reduce LDL cholesterol	• reduce triglyceride inflammation	

Based on Sources :

Animal fats

animal fats come from various sources including:

<u>fats</u>	<u>Sources</u>
meat	beef, pork, poultry
dairy	milk, butter, ghee
lard	pig fat
tallow	beef or mutton fat

Vegetable oils

Vegetable oils are derived from a variety of plant sources. It is extracted from plant seeds, fruits.

<u>oil</u>	<u>Sources</u>
olive oil	olive
sunflower oil	sunflower seeds
soybean oil	rapeseed seed
palm oil	palm fruits

(- colorless)

ProNotes

Date _____

Marine oil

Marine oils primarily come from fish, Krill and algae.

<u>oils</u>	<u>sources</u>
Fish oils	salmon, tuna, herring
krill oils	krill
Algal oil	schizochytrium
Cephalopod	squid

essential oil

essential oils are highly concentrated extracts from different plants that are used in aromatherapy.

Tea tree oil	tree leaves
lemon oil	lemon peel
Rose oil	rose petals
Jasmine oil	Jasmine flowers

Difference Vegetable and essential oil

- | | |
|--|--|
| (*) Almost odorless | (*) strong odor |
| (*) It is extracted from oleaginous fruits | (*) extracted from plants with many aromas |
| (*) Medicine cosmetic use | (*) only therapeutic use |

Based on Physical Properties :

(a) Fats have high melting point (saturated)

(b) Oils have low melting point (unsaturated fatty acid)

(c) Insoluble in water

(d) Soluble in organic solvents i.e., alcohol

(e) When heated to temperature above 300°C , fats and oil decomposes to give irritating fumes

Based on Chemical properties:

Hydrogenation of fats:

Hydrogenation of oils is known as hardening of oils because it produces solid fats. The process involves passing hydrogen into an unsaturated oil at about 200°C and 5 atm.

(c) unsaturated oil $\rightarrow$ saturated oil $\rightarrow$ hardened

Alkaline hydrolysis / formation of soap:

Saponification is the reaction between an ester (fats/oils) and an alkali to form NaOH and glycerol.

Acid hydrolysis of fats and oil:

When natural fats or oils are boiled with water in the presence of concentrated H_2SO_4 . Probiotes

Date: _____

Classification of vegetable oil

Introduction : The discovery and use of vegetable oil back thousands of years, with evidence of olive oil production in ancient Greece around 3000 BCE.

Other ancient civilisations such as the Egyptians, Indians and Chinese also used various plant based oils for cooking, medicine and rituals.

Definition : Vegetables oils can be define as :

- triglyceride extracted from a plant
- any of group of oils that are ester of fatty acids and glycerol and are obtained from plants
- Oil extracted from seed or other part of plants and used in food products and industrially.

Kings Net

Date:

Example : Soybean oil, grape seed oil and cocoa butter are example of seeds oil or fat from seeds.

Olive oil, palm oil and wheat bran oil are examples of fat from other part of plant.

Classification of vegetable oil : Vegetables oil can be classified based on :

• Sources

• Processing method

• Nutritional content

• Usage

• Chemical composition (Degree of unsaturation)

• Other classifications

• Quality grades

Classification of vegetable oil on basis of their source.

Vegetable oil can be classified based on their source into:

- 1- Seeds oil: Seeds oil are vegetable oil extracted from seed of various plants.
i.e., Sunflower oil from sunflower seeds,
Sesame oil from sesame seeds,
Pumpkin seed oil from pumpkin seeds.
- Seed oils can be rich in various nutrients such as essential fatty acids, vitamins and antioxidants.
- They are commonly used in cooking, frying, salad dressings and food processing.

Fruits oil: Fruits oil are vegetable oil extracted from fruit of various plants.
i.e., Olive oil from olives, palm oil from palm fruit, Avocado oil from avocados.
They are rich in monounsaturated, polyunsaturated fats, vitamins and antioxidants.



3. Nuts oil : Nuts are vegetable oils extracted from nuts i.e., Peanut oil from peanuts, Almond oil from almonds & Hazelnut oil from Hazelnuts.

→ Nuts oil are rich source of antioxidants, Vitamin E etc.

→ Nuts oils are mostly used in cosmetics and personal care products, flavouring dishes and salad dressings etc.

Grain oils : Vegetable oils extracted from grains i.e., corn oil from corn germ, rice bran oil from rice bran.

They are rich in sterols, antioxidants and polyunsaturated fats.

They are used in food industry (Food processing):



Date: _____

Classification of vegetable oil on basis of their processing method.

1- Cold Pressed oils: Oils are extracted from their source using a mechanical press without heat or chemicals.

I.e. Extra virgin olive oil, cold-pressed sunflower oil & sesame oil.

• Benefits of cold pressed oils: This process helps to preserve natural flavour and aroma as it helps to retain oils distinct taste and smell & nutrients and quality due to minimal processing.

• Cold pressed oils are often preferred for cooking, dressings, health supplements etc.

2- Expeller Pressed oils: Oils are extracted using a mechanical press generating heat but without chemical solvents.

I.e. Canola oil, sunflower oil etc. Kings Pikes



→ Benefits of expeller pressed oil : It offers less expenses than cold presser and extract more oil (higher yield + cost effectiveness, affordability) .
It also retains good nutrients and flavours.

- Solvent Extraction oil : Solvent extracted oils are obtained using chemical solvents such as hexane to extract oil from seeds, nuts or grains. i.e.: Soyabean oil and corn oil.

Solvent extracted oils undergo refining to remove chemical residues and impurities to improve oil quality.

This technique is widely used in industrial oil production due to its high percentage yield, efficiency and cost effectiveness.

Date: _____

Classification of vegetable oil on the basis of their nutritional content:

1. Monosaturated oils: Oils that contain a high proportion of mono-unsaturated fatty acids (MUFA). i.e. → Olive oil, Macadamia oil etc.
- These oils are often recommended as a part of a balanced diet due to their potential health benefits. as they can help
- Lower LDL (bad) cholesterol
 - Improve heart health
 - Increase HDL (good) cholesterol.

2. Polyunsaturated oils: Oils that contain a high proportion of polyunsaturated fatty acids (PUFAs). i.e. → Sunflower oil, Corn oil, Soyabean oil.
- These oils contain essential fatty acids such as omega-6 and omega-3 which is important for various bodily functions. This oil also helps to support heart health and reduce inflammation.

Date: _____

- Saturated oil : Vegetable oils that contains high proportion of saturated fatty acids. These oils are often solid at room temperature. i.e. Coconut oil, Palm oil etc.
- Some plant based oils are high in saturated fats so moderation is the key when consuming these oils as they
 - raise LDL (bad) cholesterol
 - Increase heart disease risk.

- Omega-3 rich oils : High in omega-3 fatty acid that may help to reduce inflammation and improve heart health. i.e. Flaxseed oil and chia seed oil.

- Omega-6 rich oil : High in omega-6 fatty acid may help support skin and hair health. i.e. sunflower oil, corn oil etc.

Kings Notes

Classification of vegetable oil on basis of their usage.

- 1- High-heat cooking oil: Suitable for frying and high heat cooking i.e. canola oil.
- 2- Low-heat cooking oil: Suitable for sautéing, baking and low heat cooking i.e. olive oil.
- 3- Finishing oil: Added to dishes just before serving to enhance flavour and aroma e.g., extra virgin olive oil drizzle on hummus.
- 4- Used in food processing: Vegetable oils used in production of processed foods such as snacks, baked goods, sauces etc.
- 5- Non-food applications: Vegetable oils used in cosmetics, pharmaceuticals and biofuels etc.

Classification of vegetable oils on basis of their chemical composition (degree of unsaturation).

- 1- Drying oils: These oils are known for their ability to dry and harden when exposed to air e.g., tung oil extracted from tung tree used on food contact surfaces like cutting boards.
- 2- Semi-drying oils: These oils are partially drying have intermediate properties b/w drying and non-drying oils e.g., Sunflower oil and cottonseed oil.
- 3- Non-drying oils: These oils do not readily dry or harden and have normal consistency. e.g., peanut oil, castor oil etc.

Classification of vegetable oils on basis of their quality grades.

Edible vegetable oil is used in almost all dishes so, it's highly important to know which one is good for you. Here are few basics about edible vegetable oil which will help you select right oil.

Second Grade oil : The initial product of vegetable oil extraction, containing more impurities and potentially prone to smoke during cooking. It is less refined oil in developed countries, second grade oil is often eliminated from food supply.

First Grade oil : After the absorption, decolourization, neutralization, deacidification and vacuumation of second grade oil, we get clear, transparent oil that contain fewer impurities.

3- Premium edible oil : Once again, we get premium edible vegetable oil after the absorption decolorization and vacuumation of first grade oil. It is pale yellow in colour, no oily smoke produced during cooking.

Salad oil : It is excellent in quality due to a series of strict refining processes including hydration degumming, neutralization, deacidification, absorption, decolorization, bleaching, crystallization, deumaxing and vacuumation. It is light in colour, almost no impurity, oily smoke and oily fishy smell.

Seven aspects to judge the edible vegetable oil quality.

1. **Package**: Check the package carefully to figure out whether the information is complete which include a label, brand, fat level, quality especially production and expiry date.
2. **Transparency**: Observe the edible oil in sun or light. The superior oil should be transparent not opaque. The higher brightness, the better.
3. **Colour**: Normally, the colour of the edible oil is canary yellow, yellow or claybank. To refined oil, the refined or lighter in colour and luster, the better. The dark colour indicates that the degree of refining is not high enough.
4. **Odour**: Every kind of oil has its unique odour. Put a drop of oil in the palm and rub it.

of your hand, rub hands together to heat.
then smell it. If you smell the peculiar
smell, the oil is inferior.

g- Flavours: Touch a drop of edible oil with your
fingers and taste it. If it tastes sour, bitter,
pungent, puckery, numb etc. it is obviously
inferior oil.

i- Moisture content: The edible oil with water
is opaque and easy to deteriorate in storage.
Pour oil into small clean and dry glass bottle
screw the lid on tightly, oscillate the bottle
up and down for several times. If the
oil become milky, the oil is mixed with
water.

Sediment: There is no sediment or suspended
particles in the high quality edible oil.
Heat the oil with sediment in the
it, the oil will sputter, smoke and
foam and fried food will be dark in
colour.

Kings Notes

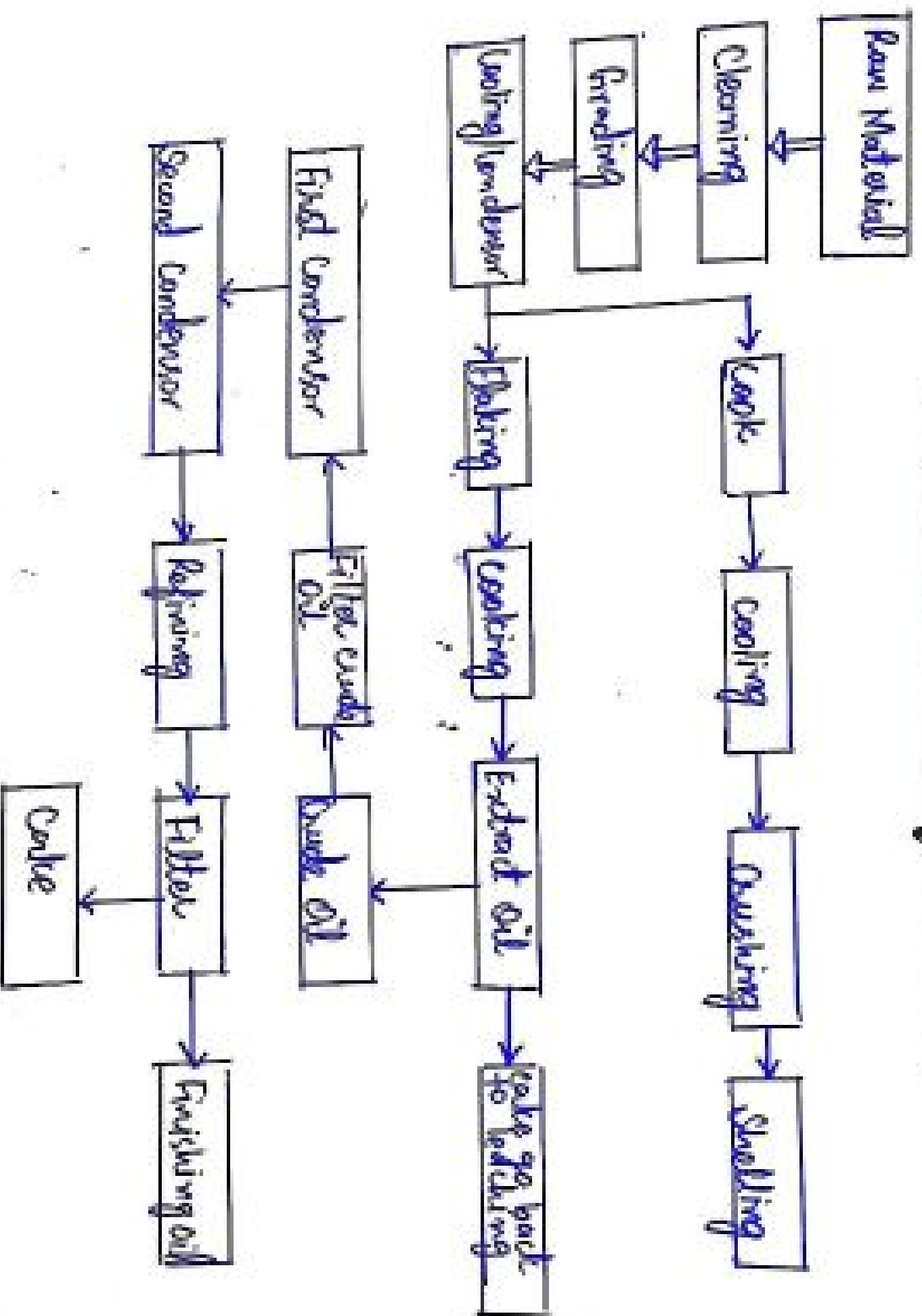
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Some oils with their uses:

1. Coconut oil: Use in soap manufacturing and cooking.
2. Cottonseed oil: A major food oil, often used in industrial food processing.
3. Olive oil: Used in cooking, cosmetics, soaps and as fuel for traditional oil lamps.
4. Palm oil: The most widely produced tropical oil also used as biofuel but majorly used in food processing.
5. Peanut oil: Mild flavour cooking oil, used as salad dressings.
6. Rapeseed oil: One of the most widely used cooking oil, canola is a variety of rapeseed.
7. Sunflower oil: A common cooking oil.

Kings Net

Extraction of oil



Raw Material

Raw materials for oil and fat extraction come from various plant and animal sources. Here are some common ones:-

i- Plant-based Raw Materials

- Seeds: Sunflower seed, rapeseed, soybean, sesame seeds.
- Nuts: Peanuts, almonds, walnuts.
- Fruits: Palm fruits, coconut, olives.
- Grains: Corn, Rice bran

ii- Animal-based Raw Materials

- Animal tissues: Lard, tallow
- Fish and Seafood: Fish oil, krill oil
- Dairy: Milk fat (butter)

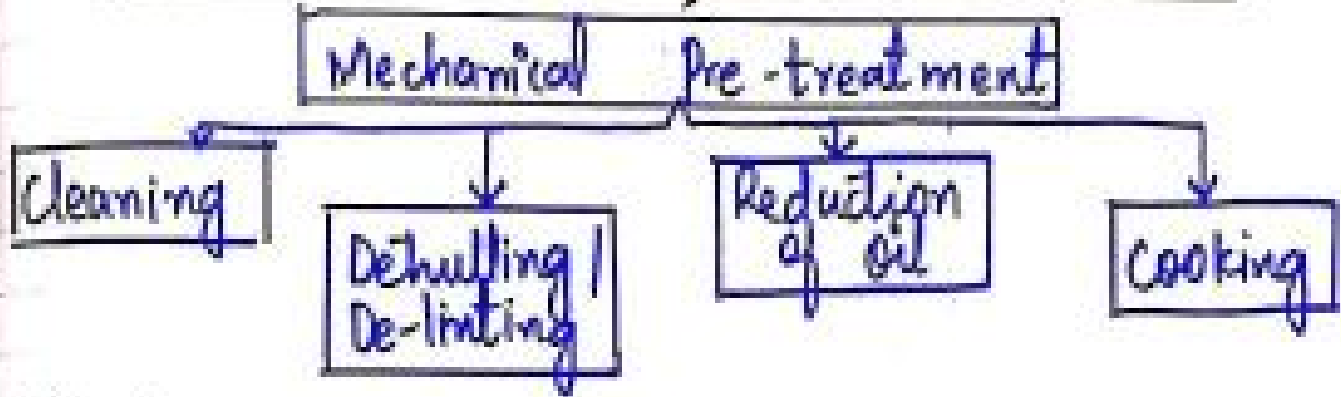
iii- Other Sources

- Micro-organisms: Algae, yeast, bacteria
- By-products: Oil extraction from food processing waste or by-products

Basic Steps:-

- 1- Mechanical pre-treatment of oil seed
- 2- Mechanical Extraction
- 3- Solvent Extraction
- 4- Refining process

1- Mechanical Pre-treatment

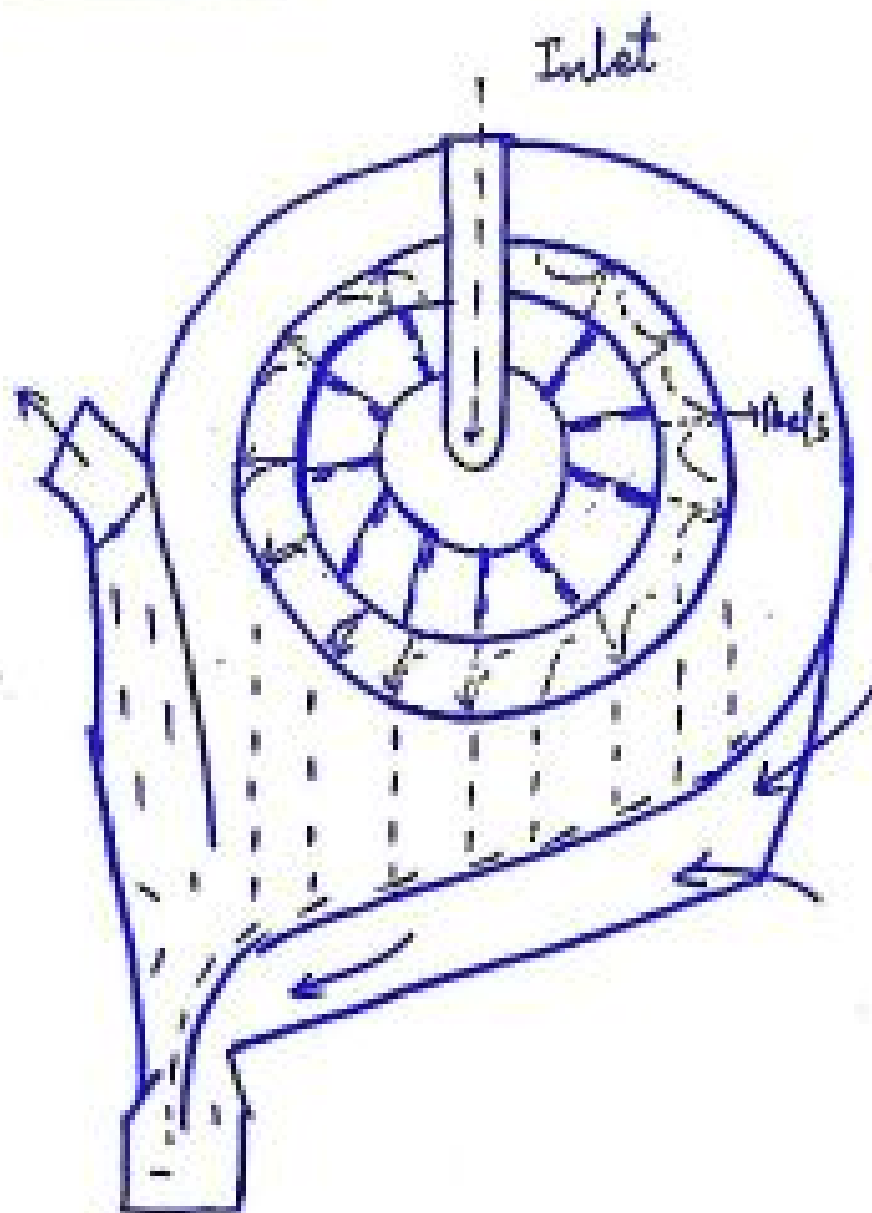


(A) Cleaning :

- Oil seed are first cleaned to separate foreign matter (eg. sticks, stems, leaves, trash) by means of revolving screens.
- Then sand or dirt is removed by screening.
- De-stoners and magnetic separator are used to remove stones and iron pieces respectively from the seed.

⇒ Cleaning of oil seeds is done before their storage to save storage space.

De-Hulling



- Aspiration
- - -> Inlet product
- - -> Peels
- - -> Kernels

(B) De-Hulling

Oil seeds are preferably de-hulled before their extraction. If the hulls are not removed before extraction they will reduce the yield of oil by absorbing and retaining oil in the press cake and in addition reduce the capacity of extraction equipment.

The hullers that usually used are:-

- Bar huller
- Disc huller

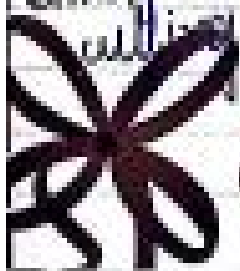
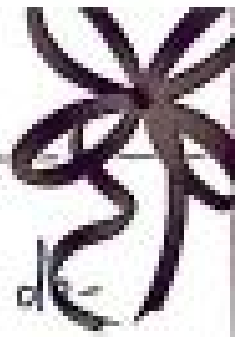
① Bar Huller :-

The seeds are fed between rotating cylinder and concave member and hulls splits as the seed are caught between opposed cutting edges.

The clearance between the cutting edges may be adjusted for seeds of different sizes.

② Disc Huller

Disc huller is more or less similar to bar huller, except that the cutting edge consist of grooves - cut on the surface of two opposed and vertically mounted disc, one



which is stationary and other rotating.
• Seed fed to the center of disc and discharged at periphery of centrifugal force.
⇒ Wet seeds are difficult to cleaning and may clog.

(C) De-Linting:-

Cotton - seeds contain lint on their surface and therefore de-linted before they are hulled.
The fibers from saw-teeth are removed by a revolving cylinder brush or by air-blast.
The lint is not removed from seeds in single operation but it taken off in 2-3 acts.

(D) Reduction of Oil Seeds

The extraction of oil from oil seeds is facilitated by reduction of seeds to small particles.

• Hammer mill, Attrition mill are used for preliminary reduction of large oil seeds such as copra, palm kernel oil.

• Normally five rollers mill is used for reduction of cottonseeds, peanuts.
• Speed: The roller move at speed

630 ft/min.

- One such mill can crush 80 tons of cotton seed in 24 hours.
- Moisture of 10-11% required in soybeans to make good flakes.

(D) - Cooking

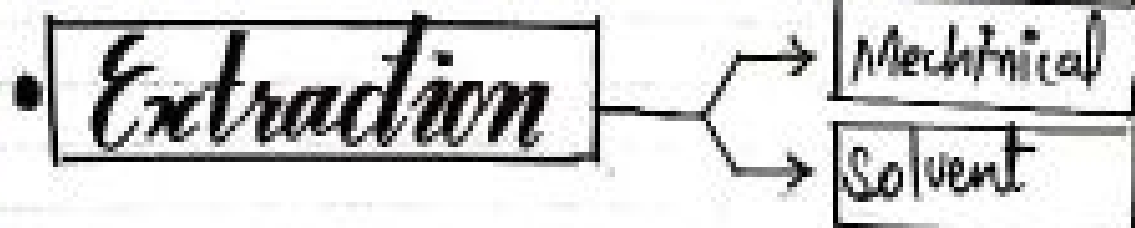
The oil droplets in cottonseed are almost "micro size", one effect of cooking cause these very fine droplets to clump into drops large enough to flow from the seeds.

Another effect of cooking is denaturation of protein.

Cooking give proper elasticity for efficient pressing, insolubilization of phosphatides, increase fluidity of oil through increase in temperature, detoxification of gossypol in cotton.

- The effect of cooking is to decrease the extraction of pigments, phosphatides and other impurities in the oil.

⇒ Over-cooking of oil seed are undesirable as it produces dark oil and cake.



2. Mechanical Extraction

(Hydraulic press / oil expeller)

Hydraulic Press

A hydraulic press extracts oil by applying mechanical force to seed or nuts, squeezing out the oil. The process typically involves preparing the material, loading it into a press chamber, and then using a hydraulic system to generate high pressure, forcing the oil through perforations into a collection container. After pressing, the solid residue (oil cake) is removed.

→ Preparation:

- The seeds or nuts are cleaned to remove impurities.
- They may be shelled and dried to ensure uniformity for pressing.
- The prepared material is then loaded into a press chamber.

→ Pressing:

- A hydraulic system, powered by a pump,

creates high pressure.

- This pressure is applied to the material in the press chamber.
- The oil is forced out through small openings in the chamber.

→ Oil Collection:

- The extracted oil flows through perforations into a collection tank.
- This tank is typically made of stainless steel for hygiene and durability.

→ Residue Removal:

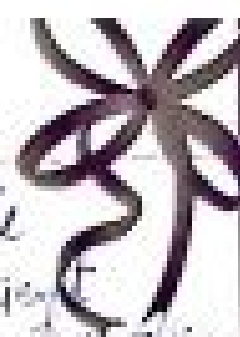
Once the pressing is complete, the remaining solid residue (oil cake) is removed from the chamber.

This residue can be used for animal feed, compost, or other applications.

→ Cycle Repeat:

- The barrel and other pressing components are cleaned for the next batch.
- The entire process can be repeated for multiple batches.

✿ "Key Features of Hydraulic press"



High Pressure: Hydraulic presses can generate very high pressures, allowing for efficient oil extraction from a variety of materials.

Controlled Pressing: The hydraulic system allows for precise control of the pressure and duration of the pressing process.

Low Heat Generation: Controlled pressure and low heat generation during pressing helps to preserve the quality of the extracted oil.

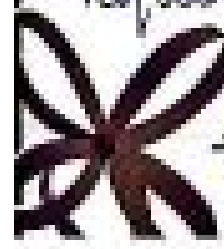
Versatility: Hydraulic presses can be used for various oilseeds and nuts.

⇒ Drawback:- (Hydraulic press)

High Maintenance: Hydraulic press require regular maintenance to ensure proper functioning.

Energy consumption: It consume significant amount of energy increasing operational costs.

Limited capacity: It may have limited capacity requiring multiple units for large scale operations.



Oil Quality: It can generate heat, potentially affecting oil quality if not controlled.

ii- Screw Press



A screw press, also known as an oil expeller, is a mechanical device used to extract oil from various oilseeds and nuts by applying pressure and forcing the oil out. This process typically involves pre-treating the seeds, then feeding them into the screw press, and finally collecting the extracted oil.

Steps: → Pre-treatment:

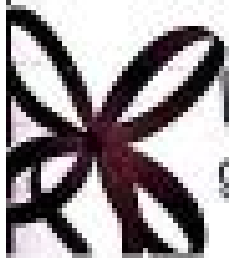
The oilseeds may be pre-treated to improve oil extraction. This can include cracking, flaking, or heating the seeds to adjust moisture and temperature.

→ Loading into press:

The prepared seeds are fed into the screw press.

→ Pressing:

The screw press utilizes a rotating screw (worm shaft) to apply pressure



Oil Extraction: - The extracted oil is typically crude and may need further refinement.

in the seeds, forcing the oil out.

→ Cake Formation

The remaining solid material (press cake) is also discharged from the press.

→ Optional refinement

The crude oil may be filtered, degummed, and further processed to improve its quality.

"Key Feature of Screw Presses"

• Continuous operations: allowing for a large volume of oilseeds to be processed at once.

• High oil yield: Screw presses can extract a significant portion of the available oil from the seeds, often achieving higher yields than hydraulic presses.

• Versatility: Screw presses can be used for a wide range of oilseeds and nuts, making them a versatile choice for oil extraction.

• Adjustment parameters:

Some screw presses offer adjustable screw speed, feed rate, and temperature processes for

different oilseeds.

Drawbacks: (Screw press)

Wear and Tear: Screw presses can experience wear and tear on the screw, barrel, and other components, leading to downtime.

Oil Quality: Screw presses can generate heat, potentially affecting oil quality if not properly controlled.

Capacity Limitation: Screw presses may have limited capacity, requiring multiple units for large-scale operation.

Maintenance Requirement: Screw presses require regular maintenance to ensure proper functioning.

Energy Consumption: Screw presses consume energy, contributing to operational costs.

3- Solvent Extraction

Mechanism

- Oil extraction is done by dissolving the seed in a solvent.
- It recovers all oil from the oilseed.
- The cake which is left behind is devoid of oil but rich in fibre and protein.
- The quality of oil is similar to that expelled oil.
- Solvent extraction plant consist of a extractor, in which oilseeds and solvent are mixed thoroughly.
- Then the oil is freed from the solvent in a series of stills and stripping columns with associated condensers.
- The oil is cooled and filtered for further storage.

Working

Solvent extraction is a method of oil extraction that utilizes a liquid solvent to dissolve oil from a solid material. This process is often used when mechanical pressing methods are insufficient to extract all the oil from the seed or other source. The most common solvent used in this process is hexane.

1-Grinding:

The oil-bearing material, such as seeds or cakes, is first ground into smaller pieces to increase surface area.

2-Solvent Immersion:

The ground material is then mixed with a solvent, such as hexanes, in a closed vessel.

3-Extraction:

The solvent dissolves the oil from the solid material, forming a mixture called miscella.

4-Separation:

The miscella is then separated from the solid residue (known as meal) through filtration or centrifugation.

5-Solvent Removal:

The solvent and mixture is then heated to evaporate the solvents leaving behind the extracted oil.

6-Solvent Recycling:

The solvent is condensed and cooled to be recycled for future extraction.

About Solvent

Ideal Solvent Properties

- Non-toxic residues in oil
- Low specific heat
- Low heat of vaporization
- Cheap
- Available in large quantity
- Non-reactive with oilseeds and extraction equipment
- Must dissolve the oil at room temperature

Solvent Use:-

- Hexane (Commonly Use)
- Pentane
- Heptane
- Octane

Alternative Solvents

While hexane is most commonly used, other solvents like petroleum ether, ethanol, and supercritical CO₂ are also being explored for solvent extraction. Supercritical fluid uses CO₂, which is a safer and more sustainable alternative.

Advantages of Solvent Extraction

- High Oil Recovery: Solvent extraction can achieve a high oil recovery rate, often exceeding 90%.
- Efficient Extraction: It's an efficient process that can extract oil from a wider range of materials, especially those with low oil content.
- Reduced Residual Oil: The resulting meal contains very little residue oil, often less than 1%.
- Versatile: It can be used for various oilseeds, oil cakes, and other materials.

Disadvantages of Solvent Extraction method

- Solvent Use: The use of solvents like hexane can pose safety and environmental concerns.

• Costs: The cost of solvents and equipment can be higher than mechanical pressing method.

• Potential Residue:

There's a risk of solvent residue in the extracted oil if not properly removed.

Asked Questions (Past Paper)

Long:-

- Describe different steps involved in the extraction of cotton seed oil. (2020, 2023)
- Write a concise note on the extraction of oil by solvent extraction method. (2019)

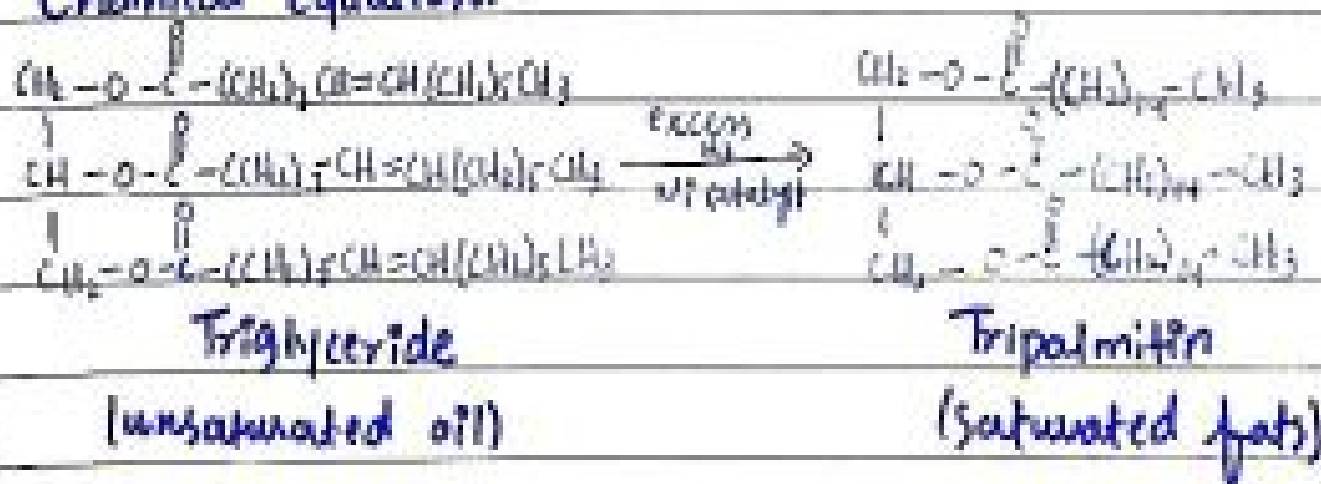
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* Hydrogenation of Oils

Definition:-

Hydrogenation of oil is the chemical process in which hydrogen gas (H_2) is added to unsaturated fatty acid in oil in the presence of metal catalyst (usually nickel) to convert them into saturated fats. This process reduces or eliminates double bond in the fatty acid chains, making the oil more solid at room temperature.

Chemical Equation:-



Kings Note:

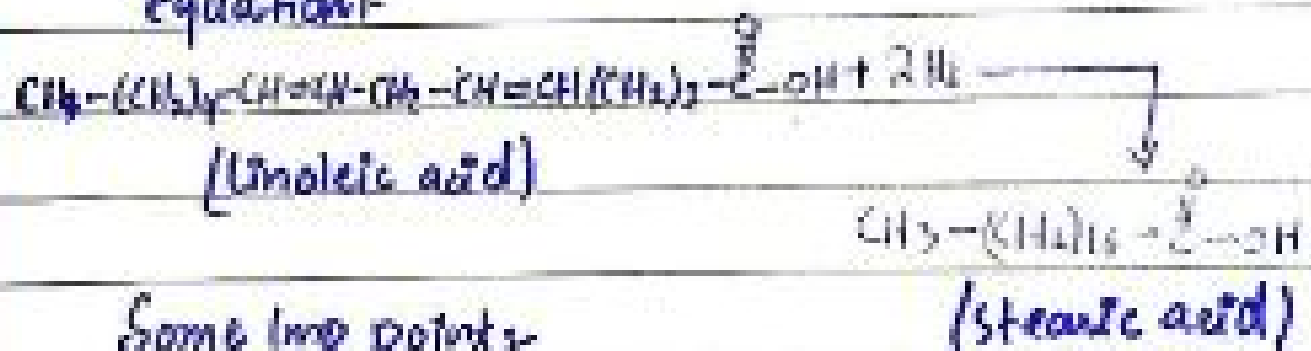
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Types of Hydrogenation

1. Complete Hydrogenation

Complete hydrogenation is a process for which all the double bonds in unsaturated fatty acid of oils are fully saturated by addition (H_2) in the presence of catalyst like nickel.

Equation:-



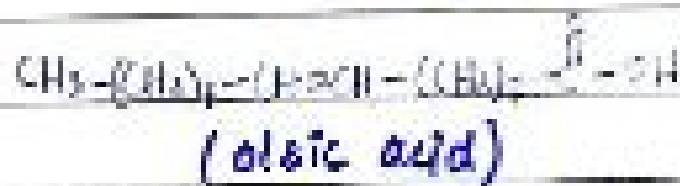
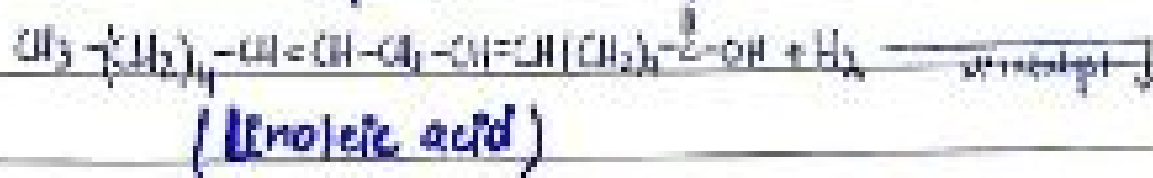
Some imp points:-

- It fully saturated
- The resulting fat is usually hard and waxy
- at room temperature
- All the double bond convert into single bond
- No trans fat is formed

2. Partial Hydrogenation-

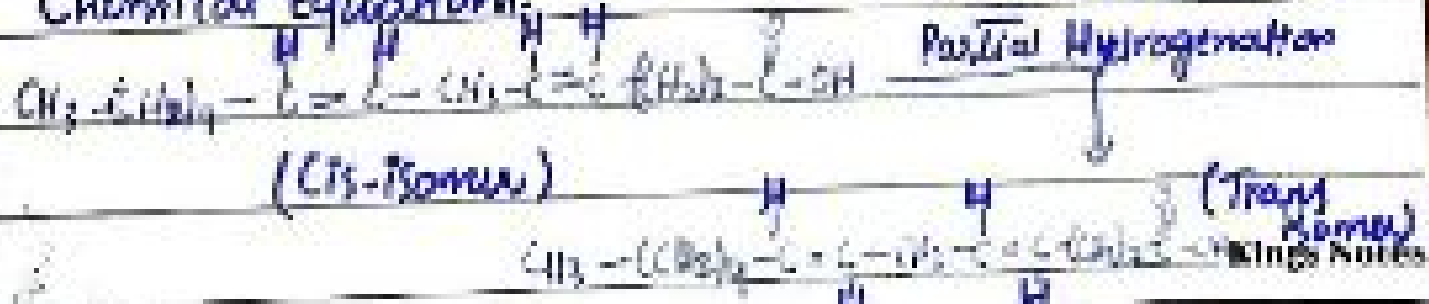
Partial Hydrogenation is a chemical process where only some of the double bonds in unsaturated fatty acid of oil are converted into single bond by adding hydrogen gas (H_2) usually in the presence of nickel catalyst.

Chemical Equation:-



Sometimes, the Cis isomer of the unsaturated fatty acid convert into trans isomer without converting the double bond into single bond form trans fats.

Chemical Equations:-



Trans Fats:- Formation and Impact

• CS Fats:-

Naturally occurring; healthy in moderation.

• Trans Fats:-

Artificial; formed during partial hydrogenation
Unnatural to humans.

Health effects

Increase LDL (Low-density lipoprotein) Bad (cholesterol).

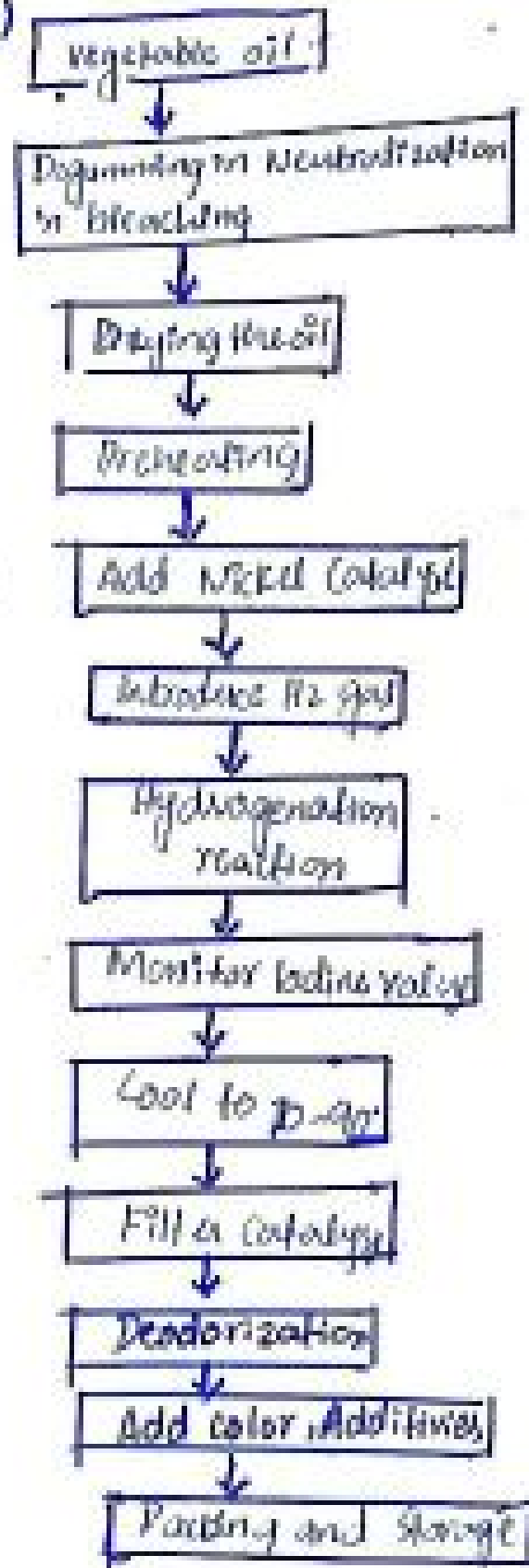
Decrease HDL (High-density lipoprotein) Good (cholesterol).

Risk factor of heart disease, stroke, diabetes.

Some Imp points:-

- May convert its double bond into trans form
- Some double bonds remain
- Lead to formation of trans fats
- Not fully saturated.

→ Flow Sheet Diagram of Hydrogenation of oil into Ghee (Fat)



Topic:

Hydrogenation of vegetable oil into Ghee (fat)

Hydrogenation of vegetable oil into ghee has four steps which are given below:

STEP-1

Oil Pre-treatment (Refining)

The crude vegetable oil may contain:

- Free fatty acids
- Gums and phospholipids
- Pigments and peroxides
- water

1. Degumming: water or acid removes phospholipids

2. Neutralization: NaOH removes free fatty acid
(Soap formation)

3. Bleaching

Bleaching clay removes pigments and traces of metals

4. Deodorization: Steam distillation removes odors

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STEP 2:- Drying the oil

Moisture is removed by vacuum drying or heating essential because water deactivates nickel catalyst. Oil must be moisture free before entering the reactor.

STEP 3:- Preheating the oil

Heat the oil to about 120-140°C in a hydrogenation reactor.

STEP 4:- catalyst Addition

Add finely powdered nickel catalyst (Raney Nickel) to catalyze the reaction. Usually suspend in oil for even distribution.

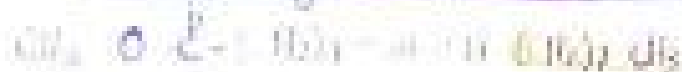
STEP 5:- Hydrogen gas is introduced

Hydrogen gas (H_2) is bubbled into the reactor when the oil is hot and the catalyst is in place.

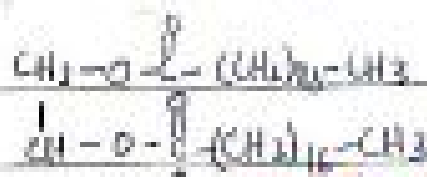
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- The temperature is raised to 180-200°C per 10 min.

The Hydrogenation reaction:-



Triolein (m.p 17°C)
(oil)



Still 2-3% of unsaturation

15-20% of unsaturation

15000 Pa

<u>Reaction Condition</u>	<u>Typical value</u>
Temperature :-	180-200°C
Pressure :-	2-5 atm
Catalyst :-	Finely powdered Nickel
Reaction Time :-	1-5 hr depending upon degree of saturation

STEP 6 :- Monitoring

- Iodine value (IV) is monitored throughout
- Measure degree of unsaturation
- Lower IV = More Saturated

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Date: _____

STEP 7:- Catalyst Removal

- After the hydrogenation reaction is complete, the oil is cooled to around 30-40°C.
- This reduces viscosity and prepares it for filtration.
- The catalyst is a solid, suspended in the liquid oil.
- It is removed by filtration by using filter press or centrifuges.

STEP 8:- Post Hydrogenation processing

→ Deodorization

Remove odors and volatile impurities using high vacuum steam distillation.

→ Additives

Color: Beta-carotene may be added for a golden yellow ghee look.

Ghee-like flavor additives may be added.

STEP 9:- Packaging and Storage

Ghee is cooled to solidify and packed into

- 1- Tins
- 2- Plastic pouches

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3- fat

Stable at room temperature after long shelf

life (6-12 months or more)

→ Application:-

1- Margarine

Used as a spread and in bakery items

Partial Hydrogenation creates desired firmness and spreadability

2- Shortenings

Used in baking for flaky textures in pastries, biscuits and cookies

3- Candle Manufacturing

Solid fats made from hydrogenated oils are used as a base for non-bleeding candles

4- Creams and lotions

• Hydrogenated oil act as emulsifiers they soften and moisturise the skin

• Common in balms, lotions, baby butter

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Date: _____

5. Soap and detergent Industry

Hydrogenated oils provide fatty acids (like stearic acid) used in soap making.

Add hardness and long-lasting lather.

Advantages of Hydrogenation Oil

1- Increase Shelf Life

- Hydrogenation reduces the number of double bonds, making oil less prone to oxidation and rancidity.
- This greatly enhances the stability and shelf life of the product.

2- Improve Texture and Consistency

- Partial hydrogenation converts liquid oils into semi-solid or solid fats (like margarine and shortening), useful in baking and cooking.
- Provides a creamy texture and good spreadability.

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Date: _____

3- Higher Melting Point:-

Saturated and trans fats formed during hydrogenation have higher melting points, making them more suitable for high temperature frying.

→ Disadvantages:-

1- Formation of Trans Fats:-

Partial hydrogenation leads to the formation of trans fatty acids.

- ▶ Increase LDL (bad cholesterol)
- ▶ Decrease HDL (good cholesterol)
- ▶ Higher risk of heart disease, stroke and type 2 diabetes

2- Loss of Nutritional Quality

Hydrogenation can destroy essential unsaturated fatty acids, reducing the overall nutritional value of the oil.

3- Health Concerns:-

Regular consumption of hydrogenated fats (especially trans fats) is associated

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with inflammation, obesity and insulin resistance

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Refining of oil

Raw oil obtained from seeds by pressing, solvent extraction, contains free fatty acids and other compounds such as phospholipids, waxes, peroxides, aldehydes and ketones, which contribute to undesirable flavour, odor and appearance, for these reasons all the oil has to be refined.

Steps involved in refining process of oil:

- Degumming
- Neutralization
- Winterization or de-waxing
- Bleaching and filtration
- Desodorization
- Polishing

1: **Degumming** Vegetable oil contains lecithins, phospholipids, and metals which are generally called because of their appearance, mallogeneous gums or simply gum. The process of elimination of the gum is called degumming. These gums are natural emulsifier, that can cause increase in viscosity which is an important parameter for the final product.

Degumming is the removal of gums from oil.

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Because of that different degumming processes are developed.

(a) **Dry degumming:** This process is generally used when low phospholipid content is present such as in coconut and palm oil. Raw oil is mixed with citric acid solution in order to coordinate metal and phospholipids. Usually the mixture is sent directly to bleaching treatment.

(b) **Soft degumming:** In this process, raw oil is treated with a water solution of chelating agent (EDTA) which coordinates with metal and phospholipids and able to remove more than 90% of phosphorus content. (e.g. expensed oil).

(c) **Total degumming process:** Raw oil is treated with acid water and then treated with base since in this step or during neutralization.

(d) **Membrane degumming:** In this process, it is used the ability of gums to form micelles which are then separated using a process similar to ultrafiltration. It also removes fatty acids. Hexane is used to form micelles.

(e) **Enzymatic degumming:** In this process enzymes, that are able to hydrolyze phospholipids, are added to raw oil. With this method it is possible to avoid use of high temperature and of acidic

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agents.

2: Neutralization:

Neutralization consists of the removal of fatty acids, which come from partial natural hydrolysis of triglyceride. In case of short fatty acids, this is done by steam stripping, such as in palm or coconut oil. Otherwise, pH is increased by adding a dilute solution, NaOH or lime. The fatty acids are transformed into soaps which are separated from the basic under phase and submitted to a second treatment with hot water at 65-90°C. The by product of this step is called soap stock.

3: Winterization or Dewaxing:

Winterization or dewaxing is a process in which oil is separated from waxes, tocopherols and residual phospholipids, which may cause turbidity issue in the oil if they remain in the final product. In a standard process, room temperature oil is slowly stirred while it slowly cooled down to cause to crystallize all the high melting temperature substance. The solids are then separated either by decantation, filtration, centrifugation or other solid liquid separation technique.

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4- Bleaching and filtration:

Raw oil contains various pigments such as chlorophylls, carotenoids, xanthophyll etc., which can cause problems with subsequent treatments as can color the final product during storage. This process aims to remove them using bleaching earth, which is a class of acidic clay that is capable of to absorb the pigments and also metals. This process can be performed after or before winterization.

Oil is mixed with this earth in 0.2-3% weight ratio, then vacuum is applied and the suspension is heated at 70-140°C to both improve the decomposition of fatty acid peroxides and the absorption of pigments. After that the oil is passed through a filter press.

5- Deodorization:

During the final steps of the refining process, the oil is stripped with vapor at a high temperature to remove all remaining undesirable odors and flavors present in it. This process removes residual fatty acids, sterols and other unsaponifiable substances. Usually, 270°C vapor is used, but an infinite temperature could be applied if the oil has

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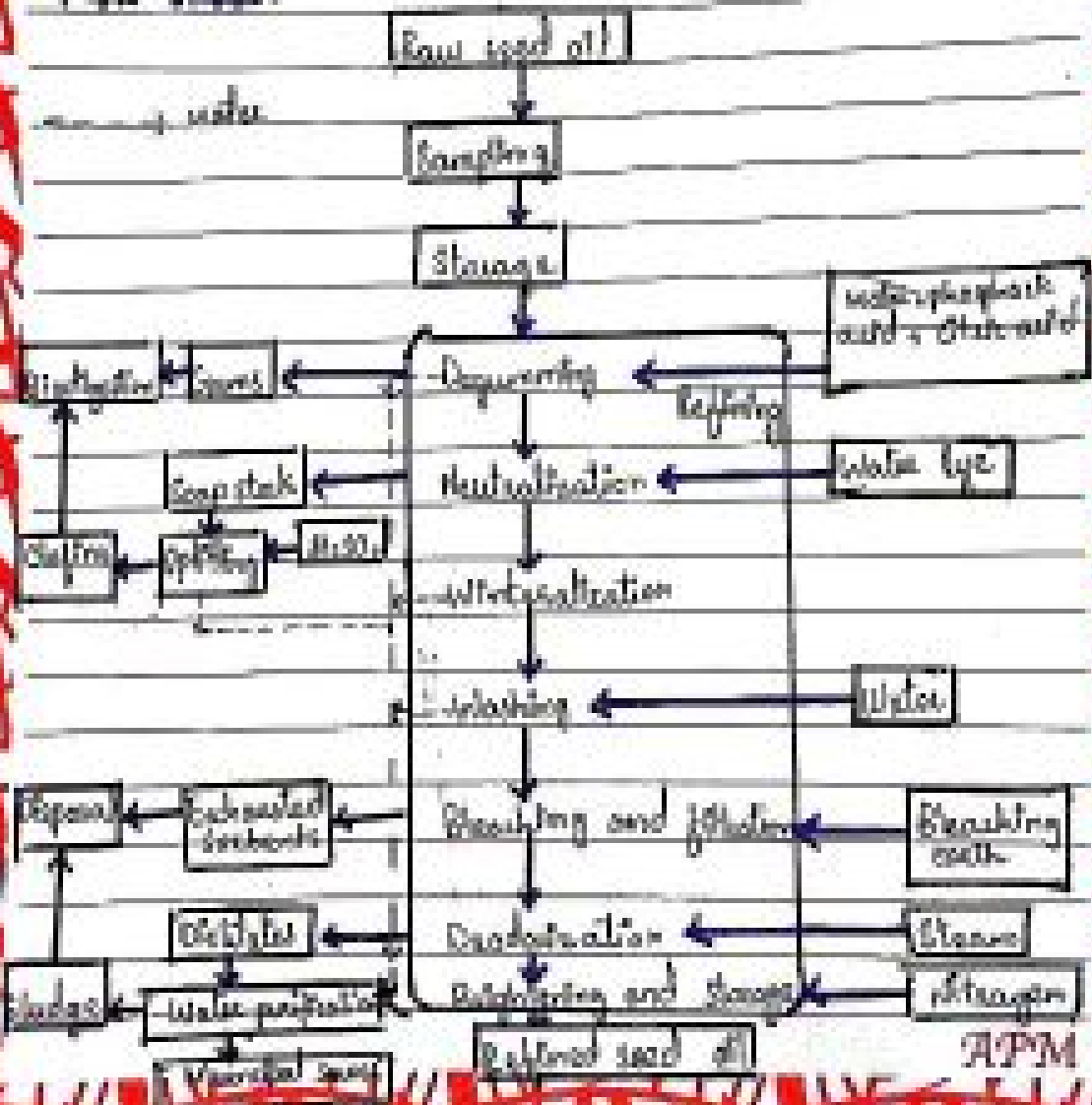
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low water content and if vacuum is applied.

6 - Polishing

As a final step before conservation, another filtration is performed. This is done to remove final residual impurities in the oil and to improve the final appearance.

Flow charts:-



"Resins"

Resins are complex, solid or semi solid amorphous substance derived from plants or synthesised chemically.

Properties:-

1. Nature

Resins are viscous liquid or solid that can be transformed into polymers.

2. Origin:-

They can be natural (plants, insects and minerals) or synthetic.

3. Solubility:-

They are generally insoluble in water but soluble in organic solvents.

4. Shape:-

They lack a regular, crystalline shape.

5. Flame:

They burn with a character smoky flame.

6. Transparent

They often appear clear and partially transparent.

7. Heavier than water

They are heavier than water.

- a. Enriched with carbon
Resins are high in carbon content.
- a. low in nitrogen and oxygen
They generally have limited nitrogen and oxygen in their structure.
- b. Variety of chemical constituents
Resins can be composed of different organic compound including terpenes, flavonoids and phenolic compounds.

Types:-

Resins come in two categories
Thermosetting:-

Thermosetting resins like epoxy and polyester, hardened when heated and cannot be reshaped.

Thermoplastic:-

Thermoplastic resins like acrylic and vinyl can be softened and reshape with heat.

Other types:-

Natural resin

Synthetic resin

Castable resin

Coating resin

Ion exchange resin

Acetal resin

Structure of resins:-

Resins structures are diverse, ranging from simple mixture of organic compounds to complex polymers.

Key features of resin structure

1. Polymeric nature

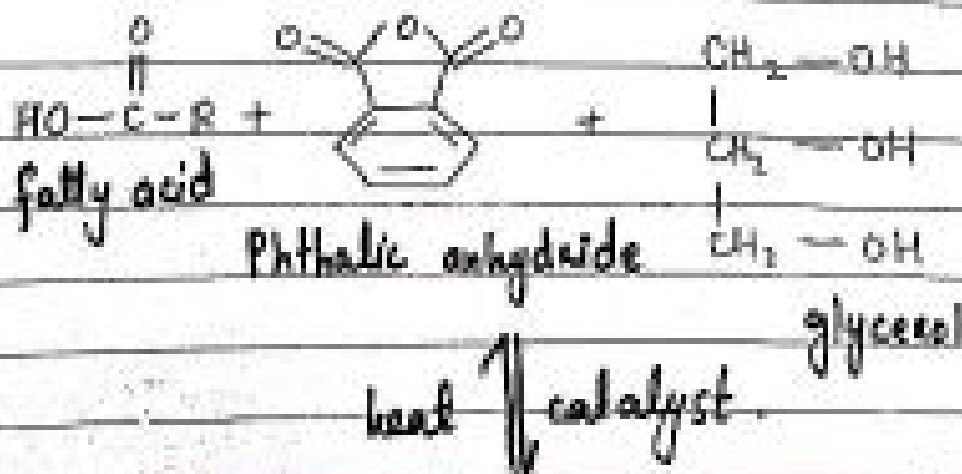
Many resins has polymeric or semi-polymeric resins

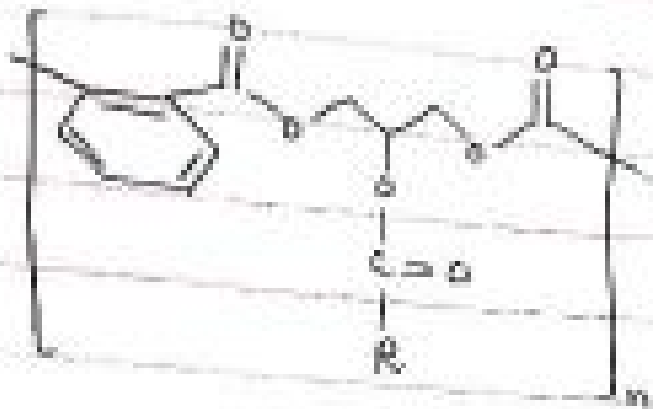
2. Cross-linking

The ability to form cross links is a key characteristics of many resins, particularly thermoplast, leading to rigid & durable material

3. Curing agent

Curing agent such as polyamines or anhydride are often used to initiate the cross linking process in resins leading to a hardened material.





alkyd resin

Importance of oil in resins:-

Oil is

crucial for resins formulations
Source of fatty acid

They contain fatty acids derived from oils especially in alkyd resins.

Influence on drying time

The more unsaturated is fatty acids, the faster the resin will dry.

Affecting flexibility hardness and durability

The type and amount of oils used can be adjusted to create resins with varying levels of flexibility, hardness and durability. For example, long oil alkyds which have

Higher oil content tend to be more flexible and durable.

Tailoring properties for specific applications
Different oils can be used to modify the properties of alkyd resins for specific application. For example, tung oil is used in paints and varnishes due to its fast drying and high durability.

Importance in coatings

Alkyd resins particularly those containing high content of oil are widely used in coatings (automobile, industries etc). Their ability to form glossy films with good weather resistance make it a valuable material.

Vegetable oil as sustainable alternative

The growing interest in sustainable material has led to development of eco-friendly alkyd resins based on vegetable oils.

Essential oils in resin art

Essential oils are sometimes used in art

for their scent and to create interesting effects such as cells in resin art.

Industrial applications of oils in resins:

Oils and fats play a critical role in the production and performance of resins, especially in the field of alkyd resins and other oil-modified resins. Here's a detailed breakdown of their applications and importance.

Alkyd resin production

Oils and fats are key raw materials in the manufacture of alkyd resins, which are widely used in paints, coatings, and varnishes.

Drying and semi-drying oils:-

Linseed oil, tung oil, soybean oil are used in air-drying coatings because they undergo oxidative polymerization

wood, protective films. It is used in decorative paints, wood finishes and printing inks.

Non-drying oils in flexible resins:-

Caster oil (a non-drying oil) produces more flexible and short resins, used in plasticizers, adhesive and some specialty coatings.

Specialized oils in Reactive resins:-

Specialized soyabean oil is used as plasticizer and stabilizer in PVC and also in epoxy resin systems for coating and composites.

Modified urethane and polyester resins:-

Soyabean polyurethanes and polyesters incorporate fatty acid derivatives for flexibility and hydrophobicity in waterproofing.

1. To reduce the surface tension of the liquid.

2. To increase the wetting of the solid surface.

3. To increase the spreading of the liquid on the solid surface.

4. To increase the adhesion of the liquid to the solid surface.

5. To increase the solubility of the solid in the liquid.

6. To reduce the viscosity of the liquid.

PURPOSES of GOOD SURFACIANT

1. To reduce the surface tension of the liquid.
2. To increase the wetting of the solid surface.
3. To increase the spreading of the liquid on the solid surface.
4. To increase the adhesion of the liquid to the solid surface.
5. To increase the solubility of the solid in the liquid.
6. To reduce the viscosity of the liquid.
7. To help in the dispersion of the solid in the liquid.
8. To help in the emulsification of the liquid.

PROPERTIES of GOOD SURFACTANT

- 1- Amphiphilic structure: Having both hydrophobic and hydrophilic parts.
- 2- Surface-active: Able to reduce surface tension
- 3- Solubility: Soluble in water or other solvents
- 4- Foaming ability: Ability to create stable foam
- 5- Emulsification: Ability to stabilize mixtures of oil and water

TYPES of Surfactants

1- Anionic

- Have $-$ ively charged head
- Examples: Sodium lauryl sulfate or sodium laureth sulfate



2- Cationic

- Have $+$ ively charged head
- Example: Methyl sulfates

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3- Non-ionic

• Have no charge on hydrophilic head, making them stable in various pH environments. Example: Sorbitan



4- AmphoTeric

• Have both +ive and -ive charges
• Example: Betaines

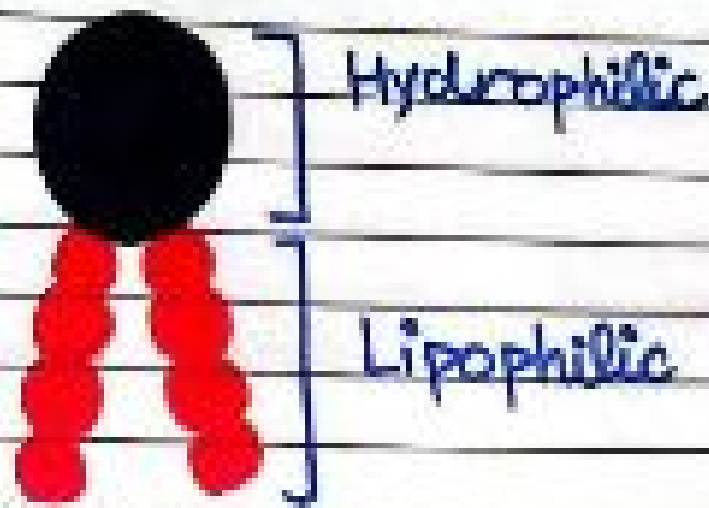


STRUCTURE of SURFACTANT

Surfactants have two parts:

- A hydrophobic tail: Water-repelling (comes from fatty acid)
- A hydrophilic head: Water-attracting added through a chemical reaction

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SAPONIFICATION-How Natural Oils Make Surfactants

- Oils provide hydrophobic (non-polar) part of surfactant molecules.
- Natural oils like coconut oil, palm oil and soybean are rich in fatty acids (lauric acid, oleic acid) which are commonly used to make surfactants.

Step 1: Natural oils are broken down to extract fatty acids like lauric acid, oleic acid, stearic acid.

Step 2: These fatty acids are then chemically reacted with:

Alkalies $\rightarrow$ ionic surfactants

Alcohols $\rightarrow$ non-ionic surfactants

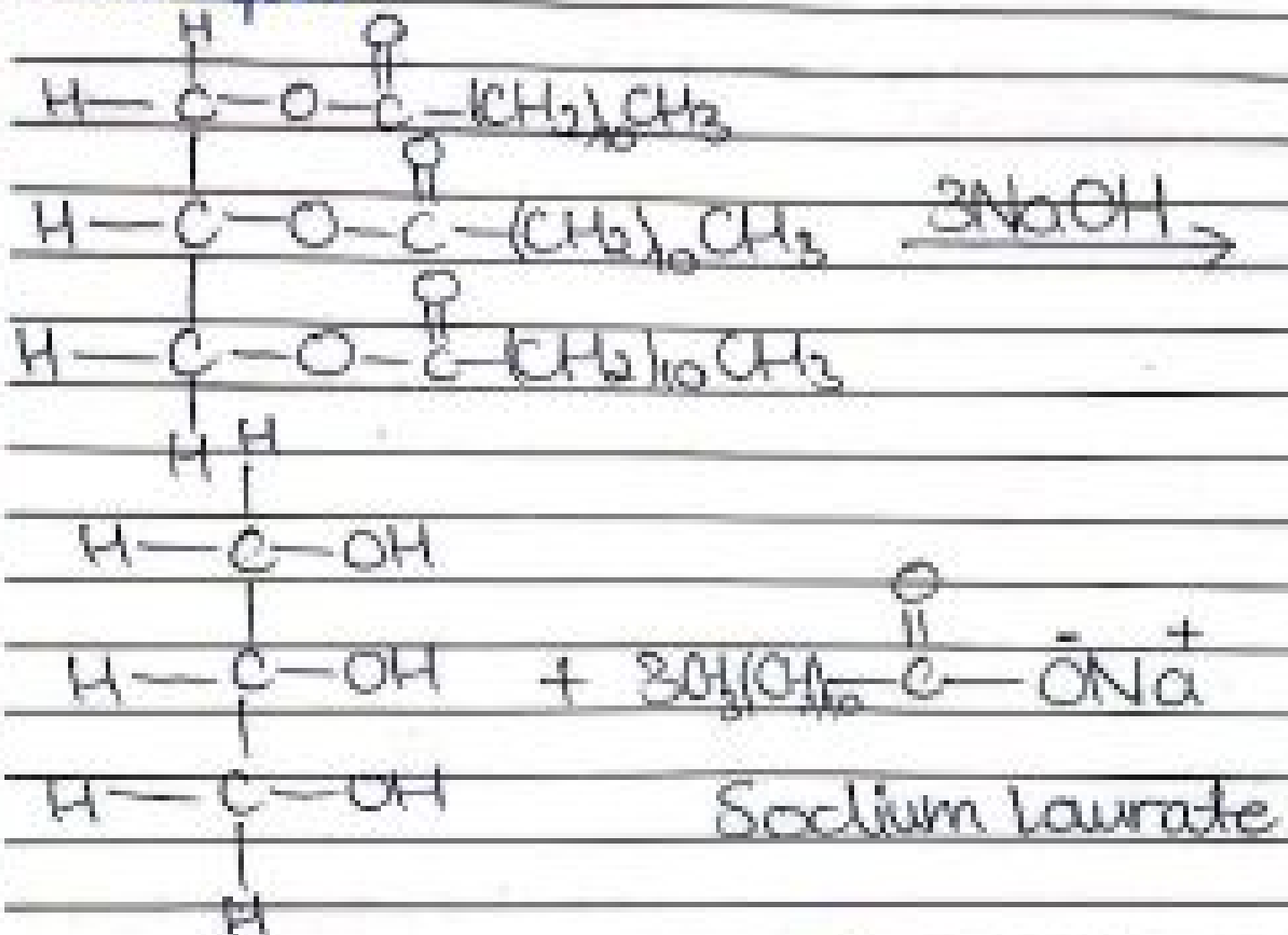
Amines $\rightarrow$ Cationic surfactants

Result: The resulting surfactant has:

$\rightarrow$ fatty acid chain as hydrophobic tail.

$\rightarrow$ hydrophilic head

Example:



ACTION of SURFACTANTS

Surfactant molecules position themselves at interface between two substances.

The tail attaches to non-polar

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part and head attaches to H_2O molecules. Hydrophobic tails disrupt hydrogen bonding between water molecules, reducing attractive forces between water molecules. Surface tension decreases. Water becomes more wetting and spreading.

Functions of OILS in SURFACIANT

- 1- Solubilization: Process of dissolving a substance (hydrophobic) in a solvent (water) with help of surfactant
- 2- Emulsification: Process of mixing two or more liquids that don't normally mix such as oil and water. Surfactants stabilize mixture, creating a stable emulsion.
- 3- Foaming: Process of creating a mixture of gas bubbles in a liquid.

Industrial Usage

- Detergents & Cleaning Products: Fatty acids from oils are used to make soaps and detergents that clean effectively by emulsifying oil & dirt.

- **Cosmetic & Personal Care:** Surfactants from oils are used in shampoo, creams and lotions to emulsify ingredients and improve texture.

- **Pharmaceuticals:** Oil based surfactants help in drug delivery systems by enhancing solubility of drugs.

- **Food industry:** Emulsifiers derived from oils help blend ingredients like oil and water to products like mayonnaise and ice cream.

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Lubricant:

Definition:

A Lubricant is a substance introduced to reduce friction between moving surfaces. It may also have the function of transporting foreign particles.

⇒ The property of reducing friction is known as Lubricity.

Characteristics of good lubricants:

A good Lubricant possesses the following characteristics:

- i- High boiling point
- ii- Low freezing point
- iii- High viscosity point
- iv- Thermal stability
- v- Corrosion prevention
- vi- High resistance to oxidation.

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Purpose:

Lubricants perform the following key functions:

- 1- Keep moving parts apart
- 2- Reduce friction
- 3- Transfer heat
- 4- Carry away contaminants and debris
- 5- Transmit power
- 6- Protect against wear.

Lubricant's Composition:

Lubricants contain 90% base oil and less than 10% additives. Typically, lubricants mainly contain oils. Oil has major application in manufacturing of oil lubricants. Although lubricants are based on one base oil, mixture of base oil are also used to meet performance requirements.

Base Oil Groups:

Mineral oil term is used to encompass lubricating base oil derived from crude oil.

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The American Petroleum Institute designates several types of Lubricant base Oil.

Group - I:

Manufacturing by solvent extraction, catalytic dewaxing and hydro-finishing process.

Group - II:

Manufacturing by hydrosacking. These base oil has superior anti-oxidation properties.

→ Synthetic Oils:

Some synthetic oils are also used in manufacturing of Lubricants.

- 1- Polyalpha-olefin (PAO).
- 2- Synthetic esters.
- 3- Polyalkylene glycols.
- 4- Alkylated naphthalenes.
- 5- Silicates esters.

ALP Notes

Application of Oils in Lubricants

Oils are a crucial component in lubricants, serving various application across industries. Here I see the main application of oils in lubricants:

(1) Base Stock for Lubricants:

Mineral Oils:

Derived from refining crude oil, they are the most common base for conventional lubricants.

Synthetic Oils:

Chemically engineered oils like polyalphaolefins (PAO), Esters and others, offering better performance in extreme conditions.

Biodegradable Oils:

Derived from vegetable sources, these are used in environmentally sensitive applications.

(2)-

Friction Reduction:

When two surfaces move against each other (like gears or engine parts), friction occurs. This friction can cause wear, heat and energy loss. Lubricating oils create a slippery thin film between these surfaces, reducing direct contact. This minimizes wear and tear, extends equipment life and ensures smoother operation.

(3)- Heat Dissipation:

How it works:

Machines and engines produce heat during operation. Oils absorb some of this heat and carry it away from hot surfaces. The heat is then dissipated when the oil circulates through coolers or is exposed to air.

Why it matters:

Overheating can lead to mechanical failure or thermal breakdown of components. Oils help to maintain a stable temperature.

4-Corrosion and Rust prevention:

Oxygen and moisture can lead to rust and corrosion, especially on metal surfaces. Lubricating oils form a protective barrier that keeps moisture and air away from metal.

Why it matters:

This prevents damage, maintains performance, and prolongs the life of equipment.

Examples:

In marine engines, corrosion resistance is vital due to constant exposure to water: special anti-corrosion oils are used.

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5- Contaminant Removal

How it works:

As machines run, tiny particles from metal surfaces, dust, or combustion residues accumulate. Lubricants oils suspend these particles and carry them to filters.

Why it matters:

It keeps internal surfaces clean, reduces abrasion, and prevents blockages or failure.

Example:

In automotive engines, oil filters trap metal shavings and carbon deposits carried by the engine oil.

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6-Sealing

How it works:

In engines cylinders or pumps, oils fill tiny gaps between parts (like piston rings and cylinder walls), preventing the leakage of gases or fluids.

Why it matters:

This maintains pressure and performance, and prevents contamination from external particles.

Example:

In internal combustion engines, oil helps to seal the combustion chamber so no gas escapes during compression.

1. Shock Absorption

How it works:

When machinery experiences sudden loads or impacts, the oil's viscosity (thickness) helps absorb and dampen the forces.

Why it matters:

This protects joints, bearings, and other components from damage due to shock loads or vibrations.

Examples:

Hydraulic oil in heavy machinery absorbs shock when lifting heavy loads or operating in rough conditions.

8. Energy Efficiency

How it works:

By reducing friction and preventing wear, lubricant oils help machines to run more smoothly with less resistance. This means less energy is needed to perform the same task.

Why it matters:

Energy savings reduce operational costs and improve overall efficiency of the system.

Example:

In wind turbines, specially formulated lubricants reduce friction in gears, improving energy output and lifespan.

Disadvantages of using oils excessive in lubricants:

Using excess oil in lubricants, either too much lubricant or oil that's too thick - can lead to several disadvantages. Here's a breakdown of the key issues:

(1) Increased Resistance and Energy Loss:

Too much oil can create excessive drag or resistance in moving parts. This reduces efficiency, requiring more power to operate machines or engines.

(2) Oil Foaming and Aeration:

Excess oil can trap air, forming foam. This reduces the oil's ability to lubricate effectively. Foaming oil can cause poor lubrication, increased wear, and even damage.

(3) Over heating:

Excess oil can retain more heat and increase fluid friction, raising the operating temperature. High temperatures degrade oil faster and can damage components.